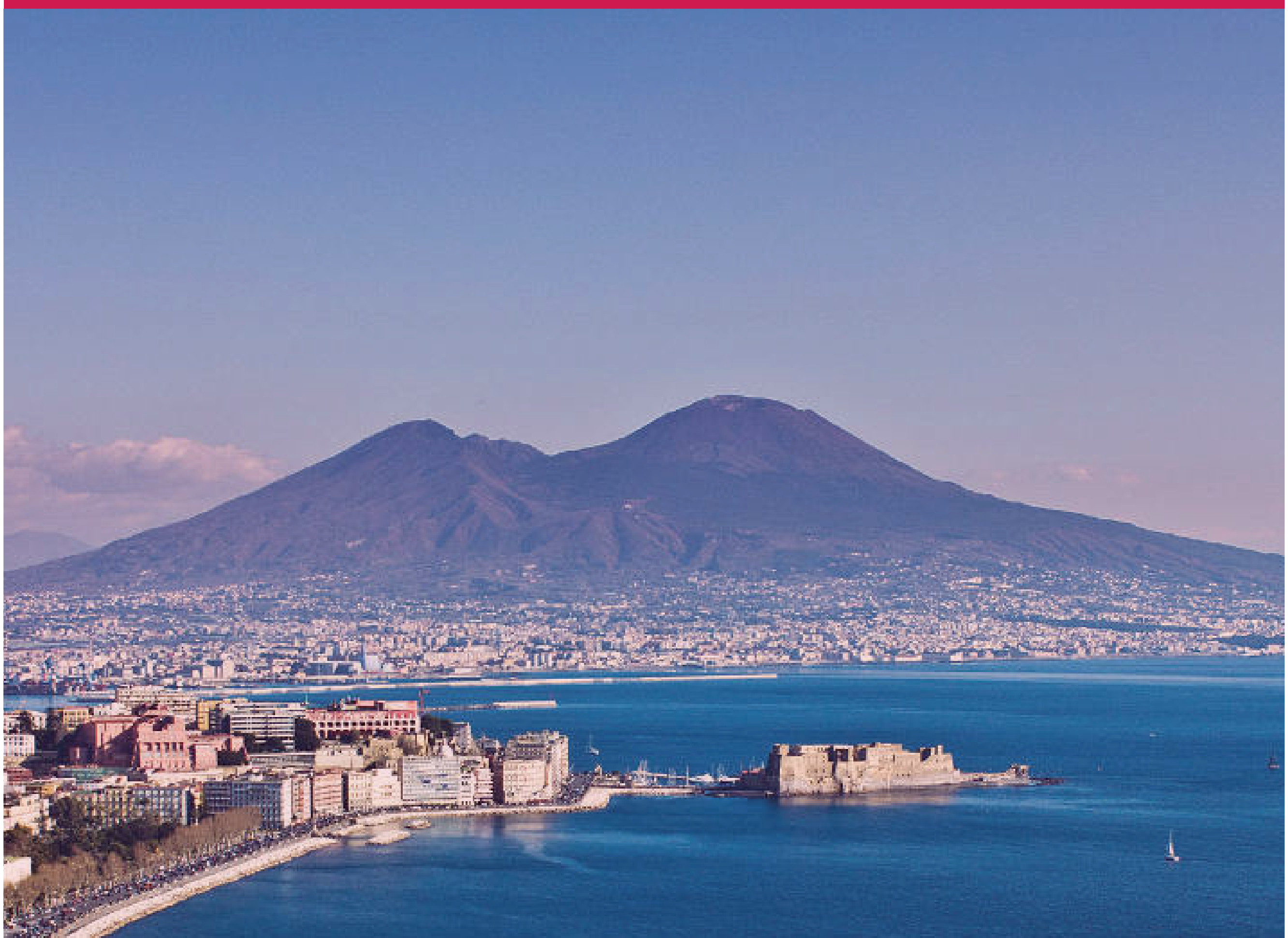
BAYESUVIUS

A visual dictionary of Bayesian Networks and Causal Inference



ROBERT R. TUCCI

Bayesuvius,
a visual dictionary of Bayesian Networks and Causal Inference

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This book is constantly being expanded and improved. To download the latest version, go to `https://github.com/rrtucci/Bayesuvius`

Bayesuvius
by Robert R. Tucci

Figure 1: View of Mount Vesuvius from Pompeii

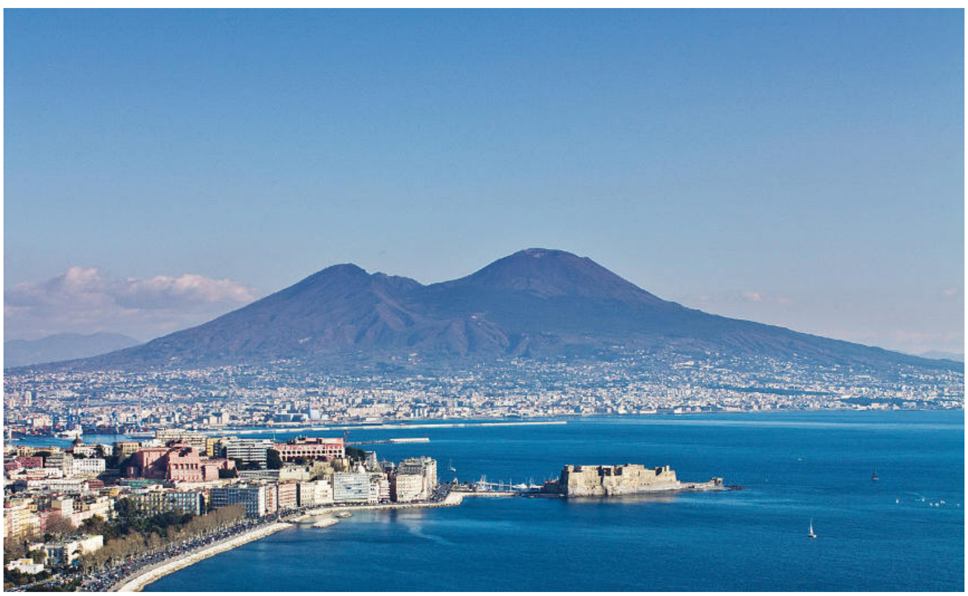

Figure 2: Mount Vesuvius and Bay of Naples

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Prologues

Chapter 50

Ising Dynamical Bayesian Network

In this chapter, we propose a dynamical Bayesian network (dbnet) (see Ch. 28) that is designed so as to incorporate all the physics of the standard Ising model (the latter is an undirected DAG (Markov model) instead of a DAG (bnet)). We have implemented our Ising-dbnet in the open source software `ising-dbnet` (see Ref.[97]).

This chapter and the software `ising-dbnet` are original research, first published here.

50.1 Preliminaries

In this section we remind the reader of some theory (on 1. Information theory 2. Standard Ising Model) that will be used to discuss the main topic of this chapter.

50.1.1 Information Theory

Recall that in Shannon Information theory (see Ch. 99), we define for random variables $\underline{x}, \underline{y}$

- the **entropy**

$$H(\underline{x}) = -\sum_{x \in val(\underline{x})} P(x) \ln P(x) \tag{50.1}$$

- the **joint entropy**

$$H(\underline{x}, \underline{y}) = -\sum_{x \in val(\underline{x})} \sum_{y \in val(\underline{y})} P(x, y) \ln P(x, y) \tag{50.2}$$

- the **conditional information**

$$\begin{aligned} H(\underline{y} \mid \underline{x}) &= -\sum_{x \in val(\underline{x})} \sum_{y \in val(\underline{y})} P(x, y) \ln P(y \mid x) & (50.3) \\ &= H(\underline{x}, \underline{y}) - H(\underline{x}) & (50.4) \end{aligned}$$

- the **mutual information (MI)**

$$H(\underline{x} : \underline{y}) = \sum_{x \in val(\underline{x})} \sum_{y \in val(\underline{y})} P(x,y) \ln \frac{P(x,y)}{P(x)P(y)} \tag{50.5}$$

$$= H(\underline{y}) - H(\underline{y} \mid \underline{x}) = H(\underline{x}) - H(\underline{x} \mid \underline{y}) \tag{50.6}$$

Here are some standard inequalities that we will find useful in the rest of this chapter.

Claim 82

$$0 \leq H(\underline{x}) \leq \ln |val(\underline{x})| \tag{50.7}$$

$$0 \leq H(\underline{y} \mid \underline{x}) \leq H(\underline{x}, \underline{y}) \tag{50.8}$$

$$0 \leq H(\underline{x} : \underline{y}) \leq \min(H(\underline{x}), H(\underline{y})) \tag{50.9}$$

proof: Left to the reader.
QED

We will use MI to measure the correlation between random variables $\underline{x}$ and $\underline{y}$.

Note that MI achieves its zero lower bound when $P(x,y) = P(x)P(y)$ (i.e., $\underline{x}$ and $\underline{y}$ are uncorrelated).

Furthermore, MI achieves its upper bound given by Eq.(50.9) if either

- $\underline{y}$ is a deterministic function of $\underline{x}$; i.e.,

$$P(y \mid x) = \mathbb{1}(\underline{y} = f(\underline{x})) \tag{50.10}$$

In that case, $\ln P(y \mid x)$ is either $0 \ln(0)$ or $\ln(1)$ so

$$H(\underline{y} \mid \underline{x}) = 0 \tag{50.11}$$

Furthermore,

$$H(\underline{y} \mid \underline{x}) = 0 \implies H(\underline{x} : \underline{y}) = H(\underline{y}) \tag{50.12}$$

- $\underline{x}$ is a deterministic function of $\underline{y}$; i.e.,

$$P(x \mid y) = \mathbb{1}(\underline{x} = g(\underline{y})) \tag{50.13}$$

In that case, $\ln P(x \mid y)$ is either $0 \ln(0)$ or $\ln(1)$ so

$$H(\underline{x} \mid \underline{y}) = 0 \tag{50.14}$$

Furthermore

$$H(\underline{x} \mid \underline{y}) = 0 \implies H(\underline{x} : \underline{y}) = H(\underline{x}) \tag{50.15}$$

Define the **MI Efficiency** $\epsilon(\underline{y} \mid \underline{x})$ by

$$\epsilon(\underline{y} \mid \underline{x}) = \frac{H(\underline{y} : \underline{x})}{H(\underline{y})} = 1 - \frac{H(\underline{y} \mid \underline{x})}{H(\underline{y})} \tag{50.16}$$

From $0 \leq H(\underline{y} : \underline{x}) \leq H(\underline{y})$ we get

$$0 \leq \epsilon(\underline{y} \mid \underline{x}) \leq 1 \tag{50.17}$$

From its definition, we see that the MI efficiency has a singularity of the type 0/0 when $H(\underline{y} \mid \underline{x}) = H(\underline{y}) = 0$.

We will henceforth refer to the $(H(\underline{y}), H(\underline{y} \mid \underline{x}))$ plane as the **entropy versus conditional info (E-CI) plane**

It is instructive (see Fig.50.1) to plot in the E-CI plane, the loci on which the efficiency equals certain values.

Another thing that I find instructive is to calculate the values of the efficiency in the case that $\underline{x}$ and $\underline{y}$ are simple binary random variables $\underline{x}, \underline{y} \in \{0,1\}$. In that case, we represent $P(y|x)$ by a 2×2 matrix, with $\underline{y} \in \{0,1\}$ labelling the rows and $\underline{x} \in \{0,1\}$ labelling the columns. Fig.50.2) gives the values of the efficiency for various $P(y|x)$ and $P(x)$ possibilities.

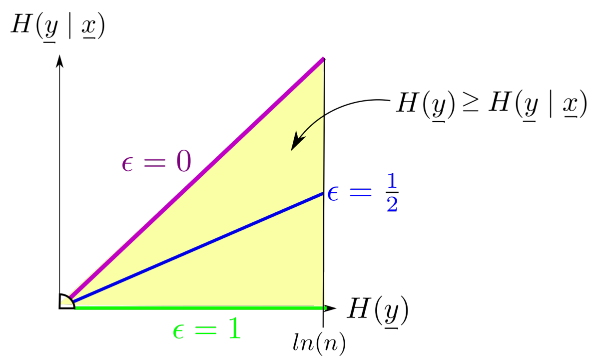

Figure 50.1: Efficiency $= \epsilon = \frac{H(\underline{y}:\underline{x})}{H(\underline{y})} = 1 - \frac{H(\underline{y}|\underline{x})}{H(\underline{y})}$. Efficiency has a singularity when both $H(\underline{y} \mid \underline{x})$ and $H(\underline{y})$ are zero.

50.1.2 Standard Ising Model

The standard Ising model is defined on a cubic lattice, as shown in Fig.50.3.

Let
$Bool = \{0,1\}$
$S_i \in \{-1,+1\} = 2Bool - 1 = \mathbb{Z}_{\pm}$
$T =$ temperature
$\beta = 1/(k_B T)$. We will set Boltzmann constant k_B to 1.
$D =$ dimension of lattice. We refer to $D = 1$ as 1-dim, and to $D = 2$ as 2-dim.

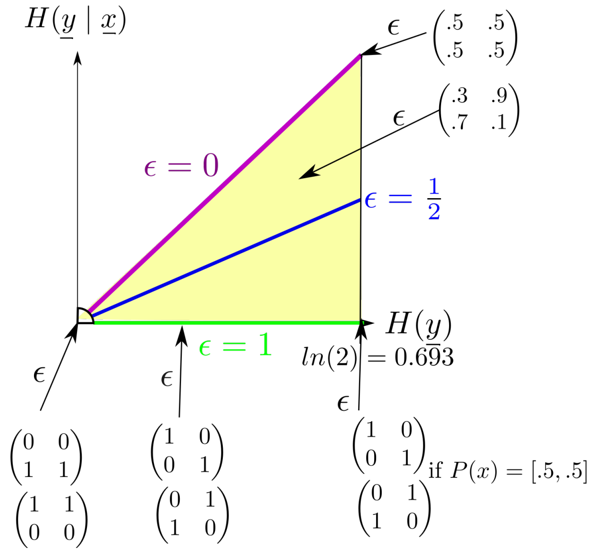

Figure 50.2: Efficiency for various values of the transition matrix $P(y \mid x)$ and the prior probabilities $P(x)$.

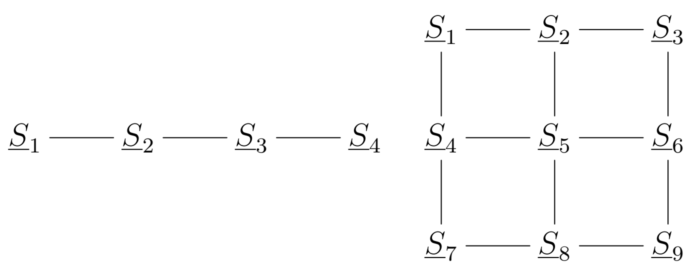

Figure 50.3: Ising Model cubic lattice for $D = 1$ (with 4 spins) and $D = 2$ (with 9 spins).

$NN(i)$ = set of indices of nearest neighbors to S_i.

$z = 2D$= **coordination number**, number of nearest neighbors internally in the lattice, away from the boundary. $z = 2$ in $D = 1$ and $z = 4$ in $D = 2$. For spins on the boundary, the number of NN is less than z. For example, as you can see in Fig.50.3, for a rectangular lattice in 2-dim, the corner sites have 2 NN, and the non-corner boundary sites have 3 NN. Only the internal sites have 4 NN.

Assume coupling constants $J, h \geq 0$.

The **D-dim Ising Model** is defined as follows.

For $i = 1, 2, 3, \ldots, n$, define **energies** $\mathcal{E}_i$ and $\mathcal{E}$ by

$$\mathcal{E}_i = \left[-JS_i \sum_{j \in NN(i)} S_j \right] - hS_i, \quad \mathcal{E} = \sum_{i=1}^{n} \mathcal{E}_i \tag{50.18}$$

and a Partition Function Z by

$$Z = \left[\prod_{i=1}^{n} \sum_{S_i \in \mathbb{Z}_\pm} \right] e^{-\beta \mathcal{E}} \tag{50.19}$$

Let

$\xi =$ **correlation length**. $\langle \underline{S}_i, \underline{S}_{i+r} \rangle \sim e^{-r/\xi}$. $\xi \to \infty$ at critical points

$\langle m \rangle = \frac{1}{n} \sum_{i=1}^{n} \underline{S}_i =$ **magnetization**

$T_c = 1/\beta_c =$ **Curie Temperature** (a.k.a. critical temperature). Magnetization is non zero for $T < T_c$ (spontaneous magnetization) and zero for $T > T_c$ (disordered phase).

The 2-dim Ising model with $h = 0$ has a critical point at $T = T_c$. If either $D = 1$ or $h \neq 0$, there are no critical points at any finite temperature, just phase transitions.

Ising Model can only be solved exactly for $D = 1, 2$. For other D, must solve numerically.

For the 2-dim Ising model with $h = 0$:

- Curie Temperature

$$T_c = \frac{2J}{\ln(1 + \sqrt{2})} \tag{50.20}$$

$$\beta_c J = \frac{1}{2} \ln(1 + \sqrt{2}) \approx 0.4407 \tag{50.21}$$

- correlation length

$$\xi \sim |T - T_c|^{-\nu}, \quad \nu = 1 \tag{50.22}$$

50.2 Scaling a General Bnet

Consider a general bnet with nodes $\underline{x}_i$ for $i = 1, 2, \ldots, n$. Let

$\mathcal{R} =$ set of root nodes.

$\mathcal{R}^c =$ set of non-root nodes.

$x^{:n} = (x_1, x_2, \ldots, x^n)$

Then

$$P(x^{:n}) = \left[\prod_{\underline{x}_i \in \mathcal{R}^c} P(x_i \mid pa(\underline{x}_i)) \right] \prod_{\underline{x}_i \in \mathcal{R}} P(x_i) \tag{50.23}$$

For $1 < b < \infty$, let

$$P_b(x_i \mid pa(\underline{x}_i)) = \frac{P(x_i \mid pa(\underline{x}_i))^b}{Z_i(b)} = \pi_i(b) \tag{50.24}$$

where

$$Z_i(b) = \sum_{x_i \in val(x_i)} P(x_i \mid pa(\underline{x}_i))^b \tag{50.25}$$

As $b \to \infty$,

$(\pi_1, \pi_2, \ldots, \pi_n)$ tends to a one hot vector; i.e., all its components tend to zero except for one component, the one that is smallest at $b = 1$, which tends to 1:

$$\pi_i(b) \to \delta(i, i_0), \quad \text{where } i_0 = \operatorname*{argmin}_i \pi_i(b = 1) \tag{50.26}$$

Therefore, $H(\underline{x}_i \mid pa(\underline{x}_i))$ (for P_b) tends to a sum of terms of the form $0 \ln(0)$ or $\ln(1)$. Thus

$$H(\underline{x}_i \mid pa(\underline{x}_i)) \to 0 \tag{50.27}$$

Thus

$$H(\underline{x}_i : pa(\underline{x}_i)) \to H(\underline{x}_i) \tag{50.28}$$

If $H(\underline{x}_i)$ remains finite as $b \to \infty$, then $\epsilon(\underline{x}_i \mid pa(\underline{x}_i)) \to 1$, but if $H(\underline{x}_i)$ tends to zero, then $\epsilon(\underline{x}_i \mid pa(\underline{x}_i))$ tends to a $0/0$ singularity at $H(\underline{y} \mid \underline{x}) = H(\underline{y}) = 0$

Claim 83

$$\sum_{i=1}^{n} H(\underline{x}_i) - H(\underline{x}^{:n}) = \sum_{\underline{x}_i \in \mathcal{R}^c} H(\underline{x}_i : pa(\underline{x}_i)) \tag{50.29}$$

Note that both sides equal zero when all the nodes are uncorrelated.

proof:

$$H(\underline{x}_i : pa(\underline{x}_i)) = \sum_{x_i} \sum_{pa(\underline{x}_i)} P(x_i, pa(\underline{x}_i)) \ln \frac{P(x_i \mid pa(\underline{x}_i))}{P(x_i)} \tag{50.30}$$

$$H(\underline{x}_i) = -\sum_{x_i} P(x_i) \ln P(x_i) \tag{50.31}$$

$$H(\underline{x}^{:n}) = -\sum_{x^{:n}} P(x^{:n}) \ln P(x^{:n}) \tag{50.32}$$

From Eq.(50.23), we get

$$\frac{P(x^{\cdot n})}{\prod_{i=1}^{n} P(x_i)} = \prod_{\underline{x}_i \in \mathcal{R}^c} \frac{P(x_i \mid pa(\underline{x}_i))}{P(x_i)} \tag{50.33}$$

Hence

$$\sum_{x^{\cdot n}} P(x^{\cdot n}) \ln \frac{P(x^{\cdot n})}{\prod_{i=1}^{n} P(x_i)} = \sum_{\underline{x}_i \in \mathcal{R}^c} H(\underline{x}_i : pa(\underline{x}_i)) \tag{50.34}$$

QED

This last claim says that the "bound entropy" (i.e., the entropy between when n random variables are independent and when they are correlated) equals the sum of the mutual informations for the non-root nodes.

Claim 84

$$\max_i H(\underline{x} : \underline{a}_i) \leq H(\underline{x} : \underline{a}_1, \underline{a}_2, \ldots, \underline{a}_m) \leq \sum_{i=1}^{m} H(\underline{x} : \underline{a}_i) \tag{50.35}$$

proof:

We will prove this for $m = 3$ and leave the general case to the reader.

Averaging the logarithm of the following chain rule for conditional probabilities

$$\frac{P(x, a_1, a_2, a_3)}{P(x)P(a_1, a_2, a_3)} = \left[\frac{P(x, a_1 \mid a_2, a_3)}{P(x \mid a_2, a_3)P(a_1 \mid a_2, a_3)}\right] \left[\frac{P(x, a_2 \mid a_3)}{P(x \mid a_3)P(a_2 \mid a_3)}\right] \frac{P(x, a_3)}{P(x)P(a_3)} \tag{50.36}$$

we get

$$H(\underline{x} : \underline{a}_1, \underline{a}_2, \underline{a}_3) = H(\underline{x} : \underline{a}_1 \mid \underline{a}_2, \underline{a}_3) + H(\underline{x} : \underline{a}_2 \mid \underline{a}_3) + H(\underline{x} : \underline{a}_3) \tag{50.37}$$

Both sides of the inequality Eq.(50.35) follow from Eq.(50.37). Indeed,

$$H(\underline{x} : \underline{a}_1, \underline{a}_2, \underline{a}_3) = \underbrace{\underbrace{H(\underline{x} : \underline{a}_1 \mid \underline{a}_2, \underline{a}_3)}_{\leq H(\underline{x}:\underline{a}_1)} + \underbrace{H(\underline{x} : \underline{a}_2 \mid \underline{a}_3)}_{\leq H(\underline{x}:\underline{a}_2)} + H(\underline{x} : \underline{a}_3)}_{\geq H(\underline{x}:\underline{a}_3)} \tag{50.38}$$

QED

Dividing Eq.(50.35) by $H(\underline{x})$ and identifying the $\underline{a}_i$ nodes with the parents $pa(\underline{x})$ of $\underline{x}$, we get

$$\max_{\underline{a} \in pa(\underline{x})} \epsilon(\underline{x} | \underline{a}) \leq \epsilon(\underline{x} \mid pa(\underline{x})) \leq \sum_{\underline{a} \in pa(\underline{x})} \epsilon(\underline{x} \mid \underline{a}) \tag{50.39}$$

50.3 Scaling our Ising-dbnet Model

So far, we have described a method of scaling any bnet. In this section, we define an Ising-dbnet and apply our scaling method to it. Our Ising-dbnet is designed to incorporate all the physics of the standard Ising model.

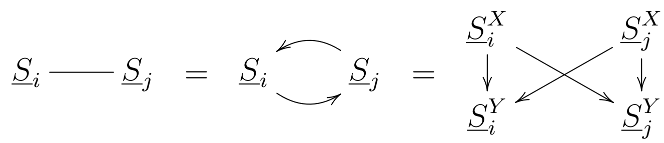

Figure 50.4: DAG definition of undirected arrow.

Let
$Bool = \{0, 1\}$
$S_i, S_i^X, S_i^Y \in \{1, -1\} = 2Bool - 1 = \mathbb{Z}_\pm$
$D_i = (S_i^X, S_i^Y)$= "dipole node" (dnode) or "lattice site"
T = temperature
$b = \beta = 1/T$

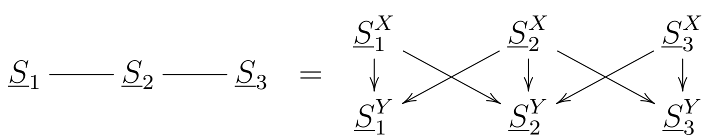

Figure 50.5: DAG definition of 1-dim Ising Model with 3 dnodes.

Assume coupling constants $J, h, \lambda \geq 0$. Actually, in our software `ising-dbnet` described later on, we only consider the case $h = \lambda = 0$.

50.3.1 1-dim Ising bnet Model

Our 1-dim Ising bnet model with 3 dnodes is defined as follows (generalization to n dnodes is obvious).

Consider the bnet of Fig.50.5 with node Transition Probability Matrices (TPM) given below, written in blue.

For $i = 1, 2, 3$,

$$P(S_i^X) = \frac{1}{2} \tag{50.40}$$

For $i = 2$,

$$\begin{cases} P(S_2^Y \mid S_2^X, S_1^X, S_3^X) = \frac{e^{-\beta\mathcal{E}_2}}{Z_2} \\ \mathcal{E}_2 = -JS_2^Y(S_1^X + S_3^X) - hS_2^Y - \lambda S_2^Y S_2^X \\ Z_2 = \sum_{S_2^Y \in \mathbb{Z}_\pm} e^{-\beta\mathcal{E}_2} \end{cases} \tag{50.41}$$

For $i = 1, 3$

$$\begin{cases} P(S_i^Y \mid S_i^X, S_2^X) = \frac{e^{-\beta\mathcal{E}_i}}{Z_i} \\ \mathcal{E}_i = -JS_i^Y(S_2^X) - hS_i^Y - \lambda S_i^Y S_i^X \\ Z_i = \sum_{S_i^Y \in \mathbb{Z}_\pm} e^{-\beta\mathcal{E}_i} \end{cases} \tag{50.42}$$

50.3.2 2-dim Ising bnet Model

Let

$\mathcal{B}$ = border dnodes of undirected graph

$\mathcal{B}^c$ = non-border, internal dnodes of undirected graph

n = number of nodes of undirected graph

Our 2-dim Ising bnet model with 9 dnodes is defined as follows (generalization to n dnodes is obvious).

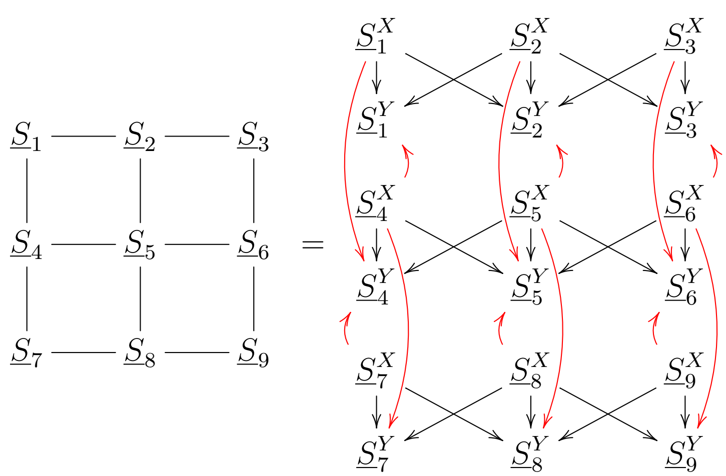

Figure 50.6: DAG definition of 2-dim Ising bnet with 9 dnodes on a 3×3 lattice.

Consider the bnet of Fig.50.6 with node Transition Probability Matrices (TPM) given below, written in blue.

For $i = 1, 2, \ldots, n$

$$P(S_i^X) = \frac{1}{2} \tag{50.43}$$

For $\underline{S}_i \in \mathcal{B}^c$, if $\underline{S}_a, \underline{S}_b, \underline{S}_c, \underline{S}_d$ are the nearest neighbors (NN) of $\underline{S}_i$,

$$\begin{cases} P(S_i^Y \mid S_i^X, S_a^X, S_b^X, S_c^X, S_d^X) = \frac{e^{-\beta\mathcal{E}_i}}{Z_i} \\ \mathcal{E}_i = -JS_i^Y(S_i^X + S_a^X + S_b^X, S_c^X + S_d^X) - hS_i^Y - \lambda S_i^Y S_i^X \\ Z_i = \sum_{S_i^Y \in \mathbb{Z}_\pm} e^{-\beta\mathcal{E}_i} \end{cases} \tag{50.44}$$

For $\underline{S}_i \in \mathcal{B}$, the corner nodes have 2 NN and the non-corner ones have 3 NN. Eq.(50.44) is sill valid for $\underline{S}_i \in \mathcal{B}$ if we set $\underline{S}_c = \underline{S}_d = 0$ in case of 2 NN and $\underline{S}_d = 0$ in case of 3 NN.

For the full bnet,

$$P(D^{:n}) = \left[\prod_{i=1}^{n} P(S_i^Y \mid pa(\underline{S}_i^Y))\right] \prod_{i=1}^{n} P(S_i^X) \tag{50.45}$$

For a 2-dim Ising bnet, Claim 83 can be specialized as follows. Using $D_i = (S_i^X, S_i^Y)$ and the fact that the S_i^X nodes are all root nodes.

$$\underbrace{\underbrace{\sum_{i=1}^{n} H(\underline{x}_i)}_{\sum_{i=1}^{n} H(\underline{D}_i)} - \underbrace{H(\underline{x}^{:n})}_{H(\underline{D}^{:n})}}_{\sum_{i=1}^{n} H(\underline{S}_i^Y) - H(\underline{S}^{Y:n})} = \underbrace{\sum_{\underline{x}_i \in \mathcal{R}^c} H(\underline{x}_i : pa(\underline{x}_i))}_{\sum_{i=1}^{n} H(\underline{S}_i^Y : pa(\underline{S}_i^Y))} \tag{50.46}$$

50.3.3 Converting 1-time bnet to a dynamical one

The 1-time Ising bnet presented so far does not equilibrate. To build a bnet that does equilibrate, we can make many copies ("times-slices") of the above 1-time bnet and connect them to make a dynamic bnet.

Add a functional dependance (t) to all the random variables $\underline{D}_i$ in the 1-time bnet presented above. Do the same with $(t+1)$. Now we have two subsequent time slices.

To connect them, set $S_i^X(t+1)$ equal to $S_i^Y(t)$ for all i. In other words, we need to add the following TPM

$$P(S_i^X(t+1) \mid S_i^Y(t)) = \quad \mathbb{1}(S_i^X(t+1) = S_i^Y(t)) \tag{50.47}$$

for all i.

50.4 Software

The ideas in this paper are implemented as free open source Python software at the github repo `ising-dbnet` (see Ref. [97])

Let

$\hat{\beta} = \beta/\beta_c$, where $\beta_c = 1/T_c$. $\hat{\beta} = 1$ at the Curie temperature T_c for a 2-dim Ising model.

In all our Jupyter notebooks, we take $J = 1$ and $h = \lambda = 0$

We consider a lattice that is 5×5. At each lattice site $i = 1, 2, \ldots, 25$, there are two nodes S_i^X and S_i^Y.

All the S_i^X nodes are root nodes of time slice t. The $S_a^X, S_b^X, S_c^X, S_d^X$ nodes of the nearest neighbors of site i point to S_i^Y

Iteration carries us from time slice t to time slice $t+1$. At the beginning of iteration $t+1$, we set $S_i^X(t+1)$ equal to $S_i^Y(t)$ for all i.

p0 is the initial ($t = 1$) probability $P(S_i^X(t = 1) = -1)$ for all i. A separate Jupyter notebook is provided for p0=0.3, 0.5, 0.7, p0=None, etc. (p0=None means each node is assigned a different p0 sampled from the uniform([0,1]) probability distribution.

We present 3 types of plots

1. **lattice plots** (see Fig.50.7)

   These show a 5×5 lattice with a color bar for efficiency ranging from 0 to 1.

   Each of the 25 lattice sites has 2 arrows connecting it to each of its nearest neighbors.

   The arrows are each labelled with the value of the efficiency for that arrow.

   Sometimes the efficiency of an arrow is undefined. In that case, the arrow is dashed.

   We use the probabilities $P(S_i^Y = \pm 1)$ to sample each node. If the sample turns out to be -1 (resp., $+1$), the node appears black (resp. white).

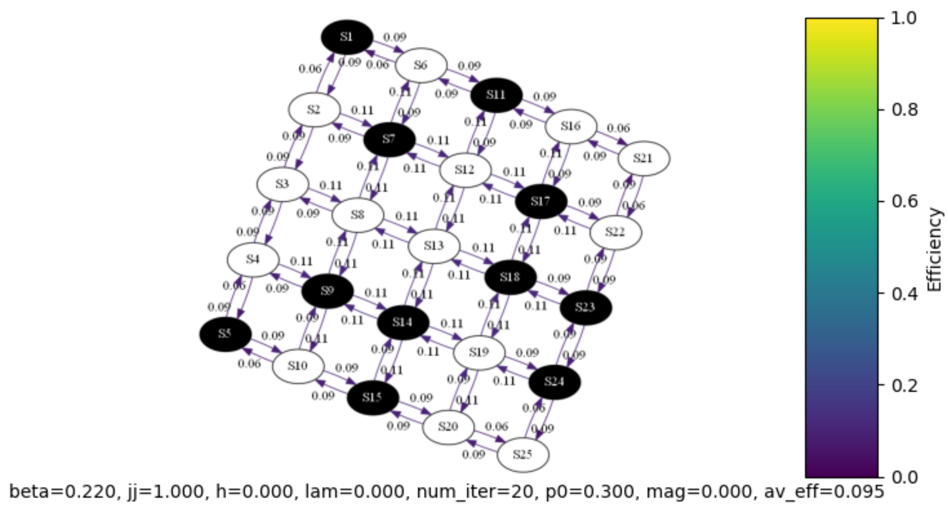

Figure 50.7: Example of lattice plot.

2. **parametric plots** (see Fig.50.8)

   These show a parametric curve in the (x, y) plane, where

   $x =$ average entropy $= \frac{1}{n}\sum_{i=1}^{n} H(S_i^Y)$

   $y =$ average conditional information $= \frac{1}{n}\sum_{i=1}^{n} H(S_i^Y \mid S_a^X, S_b^X, S_c^X, S_d^X)$

   $n = 25$

   The parameter of the curves is $\hat{\beta}$.

   The first and last point of the parametric curve are labelled by the value of the parameter $\hat{\beta}$.

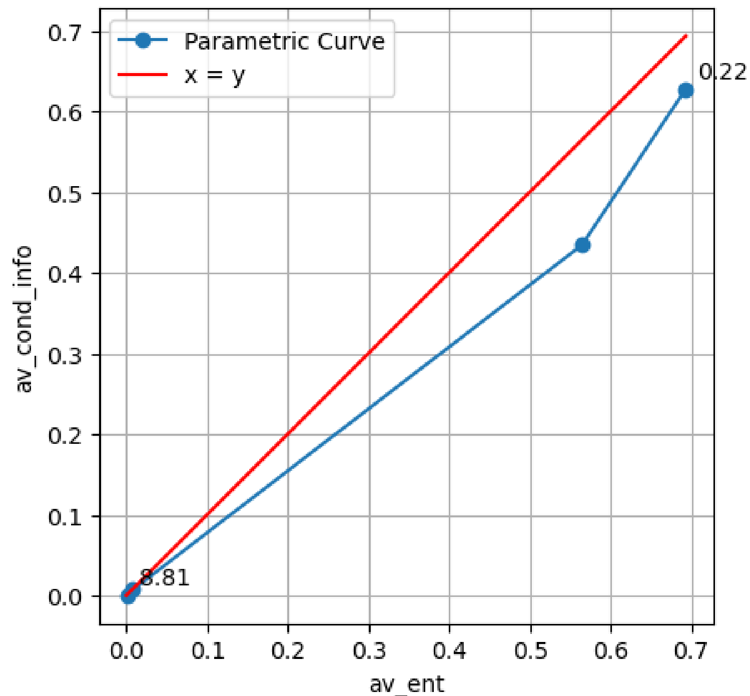

Figure 50.8: Example of parametric plot.

3. **Magnetization plots**(see Fig.50.9)

   These show the magnetization mag$=\frac{1}{n}\sum_{i=1}^{n} S_i^Y$ as a function of beta_hat$=\hat{\beta}$

Conclusions

1. Our 2-dim Ising-dbnet model exhibits a discontinuity in the magnetization at the same Curie temperature as the standard vanilla 2-dim Ising model.

2. For $p0 \neq .5$, the parametric curves flow smoothly as $\beta \to \infty$, towards $(H(y), H(y \mid x)) = (0, 0)$, where the efficiency is discontinuous.

4. For $p0 = .5$, the parametric curves flow smoothly as $\beta \to \infty$, vertically down.

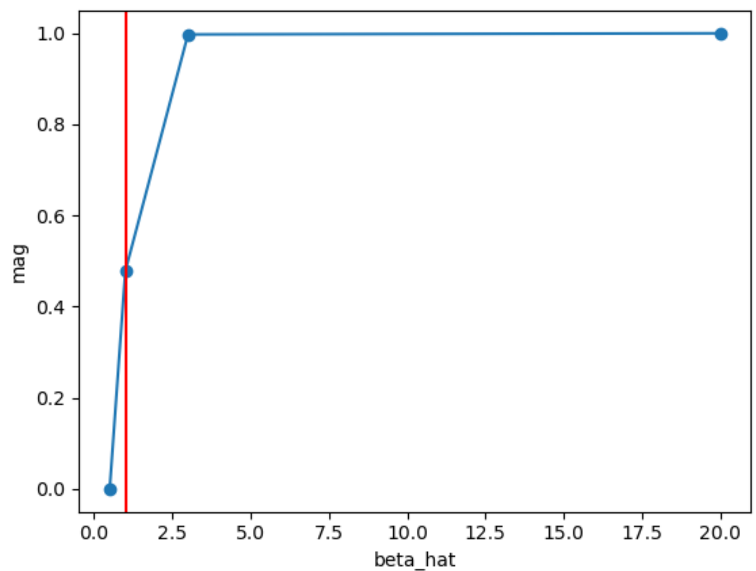

Figure 50.9: Example of magnetization plot.

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